

SALEH YAHYA

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EDUCATION

University of Illinois Urbana-Champaign

Expected Dec 2027

M.S. in Computer Science
Quant at Illinois

Lewis University

B.S. in Computer Science
B.S. in Information Technology
Teaching Assistant (TA) for Object-Oriented Programming

PROFESSIONAL EXPERIENCE

SoKat | Software Engineer Intern

July 2026 – Dec 2026

- Engineered a multi-agent LLM game platform in Python with crypto payment processing & strategy prompting for 20-round games.
- Refactored the monolith into distributed services on AWS EC2, Redis, SQS, & CloudWatch, scaling to 2k+ daily active users.
- Constructed idempotent BTC/Lightning & MetaMask escrow settling \$30k+, preventing duplicate invoices & replayed payouts.
- Integrated 6 LLM providers with OpenRouter & Fernet-encrypted keys, routing per model to shift inference cost to end users.

Paidwork | Machine Learning Engineer Intern

Sep 2025 – Nov 2025

- Improved biometric verification reliability by implementing liveness detection using camera sensors with Python & PyTorch.
- Increased document verification accuracy by refactoring object detection and optical character recognition models across multiple international identification formats, improving text extraction and layout detection by 8% using TensorFlow.
- Strengthened fraud prevention by developing anti spoofing computer vision perception models using depth estimation and texture based analysis, reducing fraudulent authentication from static image and replay attacks by 37% to a user base of 25M.

OveoAI | Software Engineer Intern

May 2025 – Aug 2025

- Migrated and transformed data pipelines from MySQL to PostgreSQL using Python, increasing database performance by 18%.
- Tuned denial model that automatically classifies denial reasons and generates corrected resubmission suggestions.
- Improved claim denial model accuracy by 12% by incorporating payer rules and coding compliance checks.

RESEARCH

MPMC Queue Performance Based Benchmarking | C++, Python, SQL | [Publication](#) | [GitHub](#)

- Benchmarked MPMC queue performance by comparing mutex synchronization against lock-free atomic implementations in C++.
- Designed and executed multi-threaded simulations across 200K–6M operations measuring throughput, latency benchmarking.
- Identified system bottlenecks including memory safety, performance degradation, and lower throughput as data and threads grew.
- Transformed data into database with Python & SQL, visualized data with dashboards, enhanced research efficiency by 20%.

PROJECTS

Luminy – Financial Data Network & Analytics Platform | Python, Scikit-learn, PostgreSQL, AWS, Redis, Nginx | luminyy.com | [GitHub](#)

- Scaled distributed fintech platform to the Top 200 in the App Store Finance category using AWS EC2, CloudWatch, and CI/CD.
- Built asynchronous institutional ingestion workflows in Python, reducing financial management processing time by 40%.
- Reduced REST API response latency by 30% through Redis caching and optimized PostgreSQL indexing strategies.
- Developed ML financial analytics model using Scikit-learn for transaction classification & time series forecasting to 1k+ users.

Limit Orderbook Matching Engine | C++, NoSQL, GTest | [GitHub](#)

- Implemented a price, time priority limit order book in C++ processing microsecond level order matching for bid and ask levels.
- Designed custom priority queues and event driven logic supporting limit, market, and cancel orders enforcing FIFO execution.
- Simulated exchange style order flow and liquidity dynamics across thousands of order events under varying market conditions.

Raycast Inspired AI Overlay | Python, Gemini API | [GitHub](#)

- Built GUI AI overlay triggered by native hotkeys for instant LLM access in Python eliminating browser tab-switching disruptions.
- Designed multi-threaded architecture separating input listener, UI thread, and worker queue for thread safe execution.
- Configured HTTP streaming to reduce perceived latency by 40% and cut 3+ manual steps per query by streamlining workflows.

SKILLS

Languages:	Python, C/C++, Java, JavaScript, SQL
Frameworks & Tools:	Kafka, Redis, WebSocket, REST API, PyTest, GTest, Matplotlib
Infrastructure:	AWS, UNIX/Linux, CUDA, PostgreSQL, NoSQL, GCC, Nginx, Jenkins, Docker, Git, GitHub Actions
Machine Learning:	PyTorch, TensorFlow, Scikit-learn, Numpy, Pandas